SERVICENOW SYSTEM ADMINISTRATOR · NAAN MUDHALVAN (SMARTBRIDGE)

Project Executable Files - Script-Controlled ACL

Phase 7 · Project Documentation

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This document lists the executable configuration artifacts for the Script-Controlled ACL project and explains exactly how to run them to reproduce the build on a fresh ServiceNow Personal Developer Instance. The project is native ServiceNow platform configuration (no external application or database), so the "executables" are a set of five numbered server-side Background Scripts plus a small amount of manual setup done through the Now Platform UI. Instance of record: dev443713.service-now.com (Australia release), built 2026-09-28.

1. Where the Scripts Run

All five scripts are server-side GlideScript and are executed from the platform's Background Scripts console. Navigate to System Definition > Scripts - Background, paste one script at a time into the editor, keep the scope on the global application, and click Run script. Each script is idempotent-friendly and should be run in numeric order. Editing Access Control rules (scripts 04 and 05) additionally requires elevating the logged-in admin to the security_admin role via the user menu (Elevate Roles).

2. The Five Background Scripts

Script	Purpose	Key tables touched	Run after
01 - Roles	Create roles bb1 (read), bb2 (create), bb3 (write), bb4 (delete) and grant all four to the test user "EEE User".	sys_user_role, sys_user_has_role	Creating the EEE User
02 - Branch choices	Populate the Branch choice field values ECE / EEE / CSE on the custom table.	sys_choice, sys_dictionary	Creating the u_institution_details table
03 - Sample records	Insert the four sample records ECE001, EEE001, EEE002, CSE001 with their branch values.	u_institution_details	Script 02
04 - ACL roles	Create the four record-level ACLs (read/create/write/delete), attach the required role to each (bb1/bb2/bb3/bb4), and remove the auto-generated u_institution_details_user role.	sys_security_acl, sys_security_acl_role	Elevating to security_admin

Script	Purpose	Key tables touched	Run after
05 - READ ACL logic	Set the READ ACL to Advanced, apply the data condition Branch is EEE (u_branch=EEE), and attach the gs.hasRole script.	sys_security_acl	Script 04

3. READ ACL Script (Advanced)

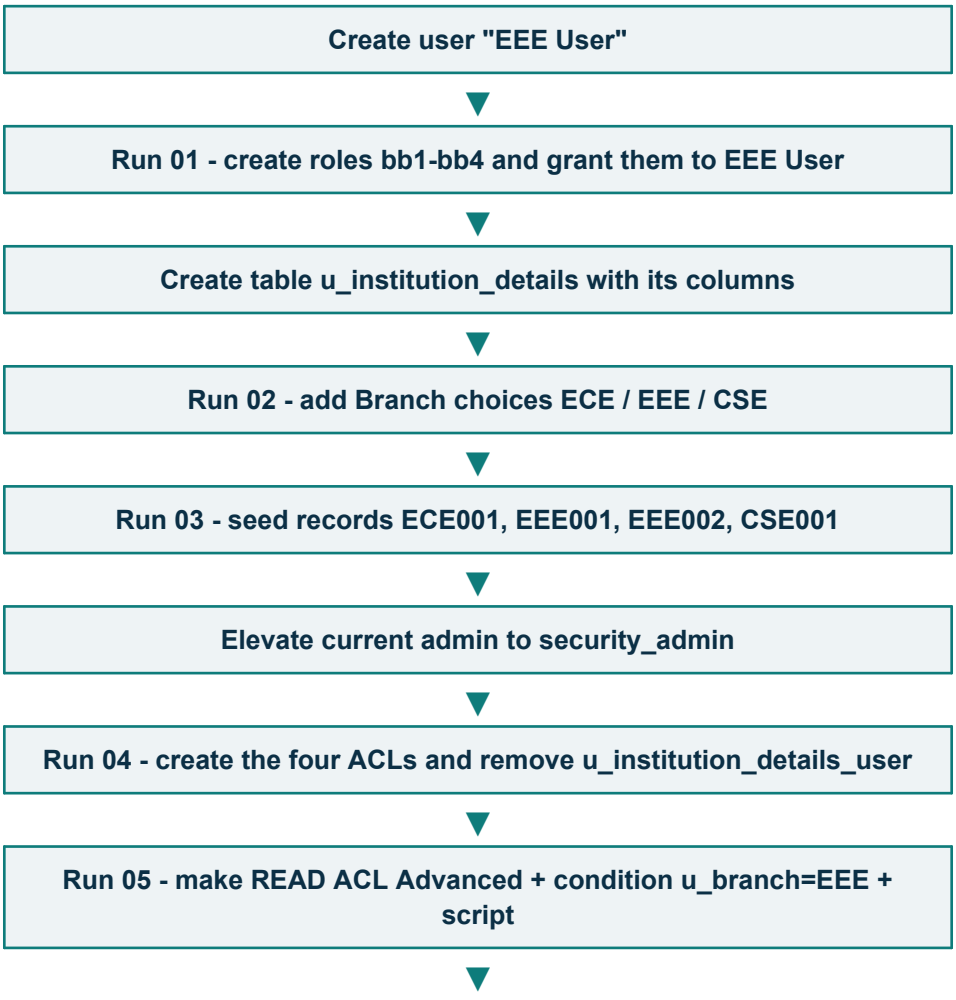
The READ ACL is the only rule that carries a script. It returns true for the platform admin and for any holder of bb1, and false for everyone else. Combined with the bb1 role requirement and the u_branch=EEE data condition, this is what limits branch users to their own records:

```
(function(){ if (gs.hasRole('admin')) { return true; } if (gs.hasRole('bb1')) { return true; } return false; })();
```

4. GitHub Repository

The five scripts are kept together as plain .js files, named 01 through 05 in run order, alongside a README that mirrors the reproduction steps below. They are version-controlled in the team's project GitHub repository maintained under the team lead's account (ArockiaRajamanickam). Because the artifacts are ServiceNow Background Scripts rather than a compiled program, cloning the repo is only a convenience for copy-paste; there is no build step and nothing to install.

5. Reproduction Steps

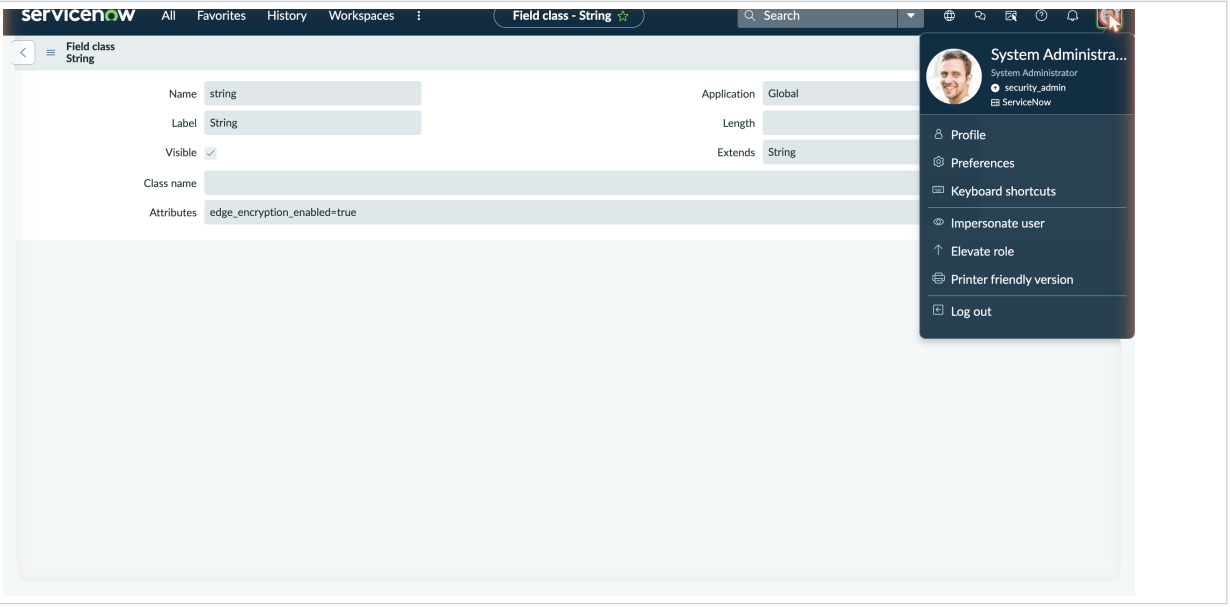


Verify by impersonation (EEE User, Abel Tuter, admin)

6. Verification Checklist

- Impersonate EEE User (holds bb1-bb4): the Institution Details list shows only EEE001 and EEE002 (2 of 2), the New button is visible, and EEE001 opens as an editable form with Update and Delete.
- Impersonate Abel Tuter (no bb roles): the list is replaced by "Security constraints prevent access to requested page".
- Return to System Administrator (admin): the list shows all four records.

Evidence (from the ServiceNow instance)



Elevated to security_admin